

1) $\boxed{100}$ NE S33 alumina

Crystal B.

$$R_F = 0.55 \text{ cm}$$

$$\text{LHR} = 300 \text{ W/cm}$$

(2) max stress of pellet?

$\frac{10}{10}$ $k_f = 0.025 \frac{\text{W}}{\text{cm} \cdot \text{K}}$ $E = 200 \text{ GPa}$

$$\nu = 0.35 \quad \alpha = 10 \times 10^{-6} \text{ K}^{-1}$$

$\sigma_{\text{hoop}} = \text{max stress}$ ✓
Fuel

$$\sigma_{\theta}(r) = -\sigma^* (1 - 3r^2)$$

$$r = \frac{r}{R_F^2}$$

$$\text{max } T_o - T_s = \frac{\text{LHR}}{4\pi k_f} = \frac{300}{4\pi \cdot 0.025}$$

$$\sigma^* = \frac{\alpha E (T_o - T_s)}{4(1-\nu)}$$

$$= 954.9 \text{ K}$$

$$\sigma^* = \frac{(10 \times 10^{-6} \text{ K}^{-1}) (200 \text{ GPa}) (954.9 \text{ K})}{4(1 - 0.35)}$$

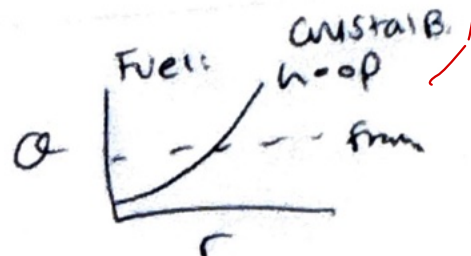
$$= 0.735$$

$$\text{GPa} = 735 \text{ MPa}$$

1a)

$$\sigma_{\theta} = -735 \text{ MPa} (1 - 3n^2)$$

$$n = \frac{r}{R_f} = \frac{R_f}{R_f} = 1$$



1470 MPa

1b)

$$\sigma_f = 175 \text{ MPa}$$

hoop stress crack

$$\sigma_f = 175 \text{ MPa} = -735 \text{ MPa} (1 - 3n^2)$$

solve for n

$$-0.238 = 1 - 3n^2$$

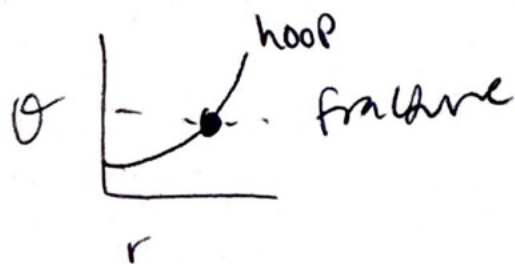
$$-1.238 = -3n^2$$

$$\sqrt{0.413} = \sqrt{n^2}$$

~~mm~~

$$n = 0.642$$

$$1 - 0.642 = 0.3576$$



35.8% of the way into the fuel is how far crack extends.

Crystal B.

2)

Clad

~~7/8~~

$$P = 15 \text{ MPa}$$

$$\bar{R}_c = 0.55 \text{ cm}$$

$$t_c = 0.05 \text{ cm}$$

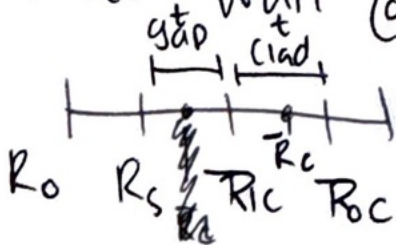
(a) thin wall:

$$\sigma_\theta = \frac{PR}{\delta} = \frac{(15 \text{ MPa})(0.55 \text{ cm})}{0.05 \text{ cm}} = 165 \text{ MPa}$$

$$\sigma_z = \frac{PR}{2\delta} = \frac{(15 \text{ MPa})(0.55 \text{ cm})}{2(0.05 \text{ cm})} = 82.5 \text{ MPa}$$

$$\sigma_r = -\frac{P}{2} = -\frac{15 \text{ MPa}}{2} = -7.5 \text{ MPa}$$

(b) Thick wall @ inside wall so R_{ic}



$$\bar{R}_c = 0.55 \text{ cm}$$

$$t_c = 0.05 \text{ cm}$$

$$\bar{R}_c = \frac{R_{ic} + R_{oc}}{2}$$

$$R_{ic} + \frac{t}{2} = \bar{R}_c$$

$$R_{oc} - \frac{t}{2} = \bar{R}_c$$

2bcont.)

Crystal B.

$$\bar{R}_c - \frac{t}{2} = R_{ic}$$

$$R_{ic} = 0.55 \text{ cm} - \frac{0.05}{2} = 0.525 \text{ cm}$$

$$R_{oc} = \bar{R}_c + \frac{t}{2} = 0.575 \text{ cm}$$

$$r = R_{ic} = 0.525 \text{ cm}$$

2b cont.)

Thick wall!

Crystal B.

$$\frac{R_o}{r} = \frac{0.575}{0.525} = 1.095$$

$$\frac{R_o}{R_i} = \frac{0.575}{0.525}$$

$$\sigma_r = -p \frac{(R_o/r)^2 - 1}{(R_o/R_i)^2 - 1} = -15 \text{ MPa} \frac{(1.095)^2 - 1}{(1.095)^2 - 1}$$

$$\sigma_r = -15 \text{ MPa}$$

$$\sigma_\theta = p \frac{(R_o/r)^2 + 1}{(R_o/R_i)^2 - 1} = 15 \text{ MPa} \frac{(1.095)^2 + 1}{(1.095)^2 - 1}$$

$$\sigma_\theta = 165.735 \text{ MPa}$$

$$\sigma_z = \frac{p}{(R_o/R_i)^2 - 1} = 15 \text{ MPa} \left(\frac{1}{(1.095)^2 - 1} \right)$$

$$\sigma_z = 75.37 \text{ MPa}$$

Crystal B.

3) calc Δ Final Δt_{gap}

16 First

14 1) $T_0 - T_s, T_s - T_{ic}, T_{ic} - T_{oc}$

2) Find T_0, T_s, T_{ic}

given $T_{oc} = 550 K$

$$T_0 - T_s = \frac{LHR}{4\pi K_f} = \frac{250 \text{ W/cm}}{4\pi (0.03)} = 663.1 K$$

$$T_s - T_{ic} = \frac{LHR}{2\pi R_f} \frac{t_g}{K_g} = \frac{250}{2\pi (0.48)} \frac{0.008 \text{ cm}}{0.003}$$

$$T_s - T_{ic} = 221 K$$

$$T_{ic} - T_{oc} = \frac{LHR}{2\pi R_f} \frac{t_c}{K_c} = \frac{250}{2\pi (0.48)} \frac{0.08}{0.15} = 44.2 K$$

3 cont.)

Crystal B.

$$T_{ic} - T_{oc} = T_{ic} - 550K = 44.2K$$

$$T_{ic} = 594.2K$$

$$T_s - T_{ic} = T_s - 594.2K = 221K$$

$$T_s = 815.2K$$

$$T_o - T_s = 663.1$$

$$T_o = 1478.3K$$

$$T_o - 815.2 = 663.1$$

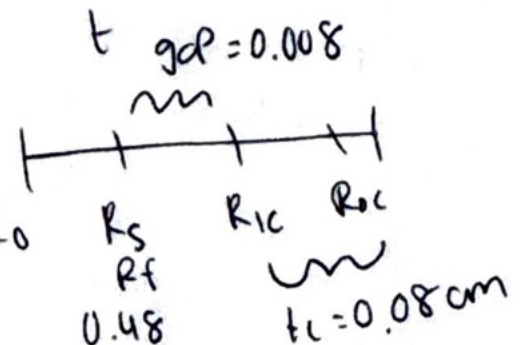
$$T_{ref} = 300K$$

$$\Delta S_{gap} = \bar{R}_c \alpha_c (\bar{T}_c - T_{ref}) - R_f \alpha_f (\bar{T}_f - T_{ref})$$

$$\bar{R}_c = \frac{R_{ic} + R_{oc}}{2}$$

$$\bar{R}_c = 0.48 \text{ cm} + 0.008 \text{ cm} + \frac{0.08 \text{ cm}}{2}$$

$$\bar{R}_c = 0.528 \text{ cm}$$



$$\bar{T}_c = \frac{T_{ic} + T_{oc}}{2} = \frac{594.2 + 550K}{2} = 572.1K$$

3 (cont.)

$$\Delta \delta_{gap} = \left[(0.528 \text{ cm}) (10 \times 10^{-6} \text{ K}^{-1}) (572.1 \text{ K} - 300 \text{ K}) \right] -$$

$$\left[(0.48 \text{ cm}) (14 \times 10^{-6} \text{ K}^{-1}) (1146.75 \text{ K} - 300 \text{ K}) \right]$$

$$\Delta \delta_{gap} = -0.00425 \text{ cm}$$

$$\bar{T}_f = \frac{T_o + T_s}{2} = \frac{815.2 + 1478.3}{2} = 1146.75$$

$$[0.0014367] - [0.00569]$$

$$= -0.00425 \text{ cm}$$

cladding

crystal B

4) hoop and rad stress @ $r=R_i$

bc thermal expansion

4/4

$$\Delta T = 50K$$

$$\alpha_c = 15 \times 10^{-6} \quad E = 100 \text{ GPa}$$

$$\nu = 0.34$$

$$t_c = 0.06 \text{ cm} \quad R_i = 0.55 \text{ cm}$$

$$r = R_i = R_o = 0.55 \text{ cm}$$

$$\sigma_r = \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left(\frac{r}{R_i} - 1 \right) \left(1 - \frac{R_i}{R_o} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_\theta = \frac{1}{2} \Delta T \frac{\alpha E}{1-\nu} \left(1 - 2 \frac{R_i}{R_o} \left(\frac{r}{R_i} - 1 \right) \right)$$

$$\sigma_r = \frac{1}{2} (50K) \frac{(15 \times 10^{-6}) (100 \text{ GPa})}{1 - 0.34} \left(\frac{0.55}{0.55} - 1 \right) \left(1 - \frac{0.55}{0.06} \left(\frac{0.55}{0.55} - 1 \right) \right)$$

$$\sigma_r = 0 \text{ GPa}$$

4 cont.)

$$\theta_\theta = \frac{1}{2} (50\text{K}) \frac{(15 \times 10^{-6})(100\text{GPa})}{1 - 0.34} \left(1 - 2 \frac{0.55}{0.06} \left(\frac{1}{-1} \right) \right)$$

$$\theta_\theta = 0.0568 \text{ GPa}$$

$$56.8 \text{ MPa}$$

5) densification

$$E_D = \Delta G_D \left(\exp \left[\frac{B \ln 0.01}{C_D B_D} \right] - 1 \right)$$

$$\beta \text{ (FIMA)} = \frac{\dot{F}}{N_D} t = \frac{(6 \times 10^{13}) (864000 \text{ sec})}{2.45 \times 10^{22}}$$

$$B = 0.00212 \text{ FIMA}$$

$$N_D = \frac{\rho N_A}{A} = \frac{(10.97 \text{ g/cm}^3) (6.022 \times 10^{23} \text{ atom/mol})}{270 \text{ g/mol}} \frac{1 \text{ U}}{1 \text{ UO}_2} = 2.45 \times 10^{22} \text{ atom/cm}^2$$

$$t = 10 \text{ day} \left(\frac{24 \text{ h}}{1 \text{ day}} \right) \left(\frac{60 \text{ min}}{1 \text{ h}} \right) \left(\frac{60 \text{ sec}}{1 \text{ min}} \right) = 864000 \text{ s}$$

$$E_D = 0.015 \left(\exp \left[\frac{(0.00212 \text{ FIMA}) \ln 0.01}{(1) (0.00526 \text{ FIMA})} \right] - 1 \right)$$

$$E_D = -0.0127 \text{ FINAL ANSWER}$$

($C_D = 1$ $T > 750$ assume)

$$B_D = \frac{5 \text{ mwd/U}}{950} = 0.00526 \text{ FIMA}$$

4/4
6) Elasticity is the ability of a mat'l to resist distortion and return to its original shape/size once the load is removed.

$$K_B = 8.1673 \times 10^{-5} \text{ eV/K}$$

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}$$

$$B_D = 5 \text{ mW/D/Kg}$$

Start: 11:03 am

End: 12:18 pm

11.23 am

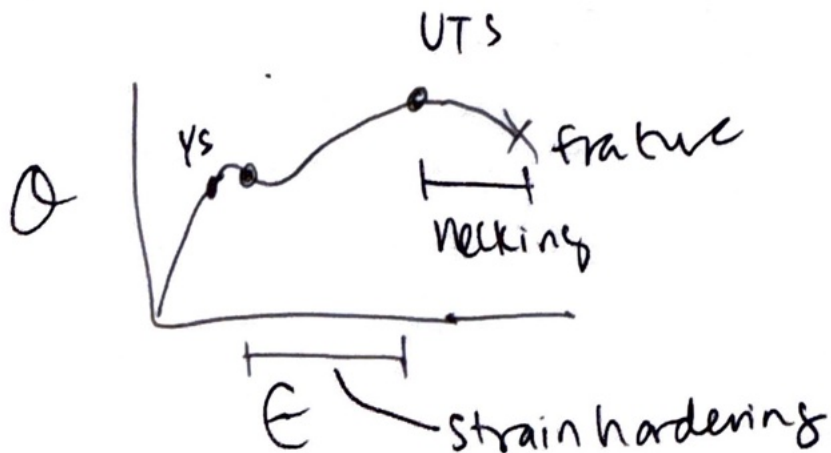
Plasticity is the ability of a material to plastically deform; non reversible change in size once load is applied.

7)

Strain hardening ~~is~~ occurs after yielding in which a material hardens and ~~up~~ increases in strength due to dislocation pileup.

Barriers cause dislocation mobility to decrease ^{and lead to pileup}, Barrier example is grain boundaries.

After the ultimate tensile strength, the material starts to neck.



Crystal

8) Fuel performance code must be able to predict fuel temperatures, specifically to centerline, predict cladding stress, and consider gap heat transfer, pressure, and gap closure.

9) Grains form in sintering process as powder is compacted at high temp. / pressure w/o melting. Particles begin contacting each other at different orientations to form grains.



10) mechanistic fuel performance modeling takes into account microstructure considerations and connects it to properties. mechanistic model relies on equations / physics rather than ~~experimental~~ experimental data. It's beneficial because unlike empirical models, you can reach ^{conditions} ~~operating regions~~ outside of what the experimental data covers and make meaningful predictions and it performs comparable and sometimes better than empirical models.

metal

11) microstructure is what
can be observed at a

7/8 magnification level of 25x

like grains. A post-processing
technique is working

such as ^{hot rolling} ~~cold rolling~~ ~~annealing~~ in which
metal is plastically deformed ~~at~~

above the recrystallization temperature
so new grains can be formed, which
is the impact on microstructure.

Related properties impact is increasing
the yield strength.

~~WMMN~~

Crystal B.

4/4 12) Diffusion occurs more quickly on the grain boundaries because there is more open space. -

6/6 13) Driving force for grain growth is reducing ~~the~~ grain boundary energy. Barriers like precipitates/
solute atoms and other defects can impede grain growth.

4/6 14) Fuel densifies during operation because fuel pellets are ~95% dense so in the reactor, the sintering process continues and small pores close (because of ^{vacancy} diffusion).
~~Driving force~~ Driving force is reduction in surface areas of pores.